Why Two Years?

Team Members: ____________ ____________ ____________

____________ ____________

Part 1 — Before we look at anything

In Lab 7 you tested for a temperature trend by counting how many times the temperature went **up** two years later versus how many times it went **down**. Here is the function you wrote:

```
def changes(array, years = 2):
    "Increases minus decreases after two years."
```

You found far more increases than decreases, got a p-value near zero, and rejected the null. Look at the `2`.

1. **What lag would you have chosen? Each team member write your own answer, then compare.**

   My lag: ________ Others on my team chose: ____________________

2. **If we ran the same test using your lag instead of 2, would the conclusion change? Circle one and give a one-sentence reason.**

   YES NO NOT SURE

____

____

STOP. Do not go on until every member of your team has answered 1 and 2.

Part 2 — Seven years, checked by hand

Below are seven consecutive years of average annual temperature. These numbers are invented, not measured — they are small enough that you can check every claim yourself.

Year	1	2	3	4	5	6	7
Temp	12.4	12.3	11.8	12.7	13.4	12.8	13.2

At a lag of 1 year, the six differences are $-0.1, -0.5, +0.9, +0.7, -0.6, +0.4$: three increases, three decreases, net 0. That row is filled in for you as a model.

3. **Complete the table for lags 2 through 6.**

Lag	Comparisons	Increases	Decreases	Net (Inc − Dec)
1	6	3	3	0
2	5			
3	4			
4	3			
5	2			
6	1			

4. **Circle the row that matches the lag Lab 7 used. Out of the comparisons it made, how many pointed toward warming?**

5. **Which lag gives the strongest evidence of warming? Is it the lag you wrote down in question 1?**

6. **Add up the Net column for all six lags. What is the total?**

   Total: ______

7. **You just combined every lag into one number. What decision did that let you avoid making?**

Part 3 — Counting what you actually have

Adding up every lag is the same thing as comparing **every year to every other year**. Check that: your Comparisons column should total $6+5+4+3+2+1=21$, and there are exactly 21 ways to pick two years out of seven. To decide whether a total like the one in question 6 is surprising, Lab 7′s approach was to build a null distribution by flipping a fair coin for each comparison — Increase or Decrease — and seeing how large the total got by chance.

8. **Suppose year 3 is colder than year 5, and year 5 is colder than year 7. Before doing any arithmetic: what must the comparison between year 3 and year 7 be?**

9. **Based on your answer to question 8, what is wrong with simulating the 21 comparisons as 21 independent coin flips — and does that error leave you too cautious about the trend, or too confident?**

If You Finish Early

A. You have 21 comparisons but fewer than 21 independent pieces of information. Roughly how many independent pieces do you think there are? Defend any number you like — there is a real answer, but the argument matters more than the number.

B. Here is a different strategy: compute a p-value separately at each of the six lags, then report the smallest one. Explain why this is not the same as adding all the lags together, even though both use all six.